Indian Institute of Science Education and Research, Kolkata
Math Methods (PH2204)
Solution of Class Test -II
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Attempt all *five* questions. Each carries 20 marks.

Notation: $z = x + iy = re^{i\theta}$, where $x = r\cos\theta, y = r\sin\theta$ and $i = \sqrt{-1}$. $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \equiv (x, y, z)$. $\nabla \equiv (\partial_x, \partial_y, \partial_z)$. Spherical polar coordinates: $\vec{r} = (r\sin\theta\cos\phi, r\sin\theta\sin\phi, r\cos\theta)$; $\nabla \equiv \left(\partial_r, \frac{1}{r}\partial_\theta, \frac{1}{r\sin\theta}\partial_\phi\right)$.

1. **(a)** Express Cauchy-Riemann condition in polar coordinates. **[5]**
   (b) Find if $f(z) = zz^* \exp(\frac{1}{z^*} - \frac{1}{z})$ is an analytic function. **[7]**
   (c) In what region of the complex plane $f(z) = |x| - i|y|$ is analytic? **[8]**

Solution:

(a) The Cauchy-Riemann condition in Cartesian coordinates is given by

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \tag{1}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \tag{2}$$

where $f(z) = u(x, y) + iv(x, y)$ with $z = x + iy$. Now in polar coordinates we have $z = e^{i\theta}$ with

$$x = r\cos\theta$$
$$y = r\sin\theta$$

Now consider

$$\begin{aligned}\frac{\partial u}{\partial r} &= \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}\\ &= \frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta\end{aligned}$$

Now using (1) and (2), the above equation implies

$$\begin{aligned}\frac{\partial u}{\partial r} &= \frac{\partial v}{\partial y}\cos\theta - \frac{\partial v}{\partial x}\sin\theta\\ &= \frac{\partial v}{\partial y}\left(\frac{1}{r}\frac{\partial y}{\partial \theta}\right) + \frac{\partial v}{\partial x}\left(\frac{1}{r}\frac{\partial x}{\partial \theta}\right)\\ &= \frac{1}{r}\frac{\partial v}{\partial \theta}\end{aligned}$$

Hence we get

$$\boxed{\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}}$$

Similarly consider

$$\begin{aligned}\frac{\partial v}{\partial r} &= \frac{\partial v}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial v}{\partial y}\frac{\partial y}{\partial r}\\ &= \frac{\partial u}{\partial x}\cos\theta + \frac{\partial u}{\partial y}\sin\theta\end{aligned}$$

Again using (1) and (2), the above equation implies

$$\begin{aligned}\frac{\partial v}{\partial r} &= -\frac{\partial u}{\partial y}\cos\theta + \frac{\partial u}{\partial x}\sin\theta\\ &= -\frac{\partial u}{\partial y}\left(\frac{1}{r}\frac{\partial y}{\partial \theta}\right) - \frac{\partial u}{\partial x}\left(\frac{1}{r}\frac{\partial x}{\partial \theta}\right)\\ &= -\frac{1}{r}\frac{\partial u}{\partial \theta}\end{aligned}$$

Hence we get

$$\boxed{\frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}}$$

(b) We have

$$\begin{aligned}
f(z) &= zz^* \exp\left(\frac{1}{z^*} - \frac{1}{z}\right) \\
&= zz^* \exp\left(\frac{z - z^*}{zz^*}\right) \\
&= |z|^2 \exp\left(\frac{z - z^*}{|z|^2}\right) \\
&= r^2 \exp\left(\frac{2i\sin\theta}{r}\right) \\
&= r^2 \left(\cos\left(\frac{2\sin\theta}{r}\right) + i\sin\left(\frac{2\sin\theta}{r}\right)\right)
\end{aligned}$$

Where $z = e^{i\theta}$. Now we check Cauchy-Riemann condition in polar form

$$\begin{aligned}
\frac{\partial u}{\partial r} &= 2r\cos\left(\frac{2\sin\theta}{r}\right) + r^2\left(\sin\left(\frac{2\sin\theta}{r}\right)\right)\frac{2\sin\theta}{r^2} \\
\frac{1}{r}\frac{\partial v}{\partial \theta} &= 2\cos\theta\left(\cos\left(\frac{2\sin\theta}{r}\right)\right) \\
\implies \frac{\partial u}{\partial r} &\neq \frac{1}{r}\frac{\partial v}{\partial \theta}
\end{aligned}$$

Cauchy-Riemann condition is not satisfied, hence $f(z)$ is not analytic.

(c) We have $f(z) = |x| - i|y|$. Checking the Cauchy-Riemann condition

$$\begin{aligned}
\frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}|x| = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases} \\
\frac{\partial v}{\partial y} &= -\frac{\partial}{\partial y}|y| = \begin{cases} -1, & y > 0 \\ 1, & y < 0 \end{cases}
\end{aligned}$$

This is satisfied in the regions $(x < 0, y > 0)$ and $(x > 0, y < 0)$. Also note that

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = 0$$

Hence $f(z)$ is analytic in the regions $(x < 0, y > 0)$ and $(x > 0, y < 0)$.

2. **(a)** Find the real and imaginary part of i^i and $\log(-1)$. **[6]**
(b) If $z = e^{i\theta}$, then show $z^n + \frac{1}{z^n} = 2\cos(n\theta)$. Use this to show $4\cos^3(\theta) = \cos(3\theta) + 3\cos(\theta)$. **[6+8]**

Solution:
(a) We have

$$i^i = (e^{i\frac{\pi}{2}})^i = e^{-\frac{\pi}{2}}$$

Hence $\Re(i^i) = e^{-\frac{\pi}{2}}$ and $\Im(i^i) = 0$. We also have

$$\log(-1) = \log(e^{in\pi}) = in\pi$$

where n is an odd integer. Hence $\Re(\log(-1)) = 0$ and $\Im(\log(-1)) = -n\pi$.

(b) We have

$$\begin{aligned}
z^n + \frac{1}{z^n} &= e^{in\theta} + e^{-in\theta} \\
&= \cos(n\theta) + i\sin(n\theta) + \cos(n\theta) - i\sin(n\theta) \\
&= 2\cos(n\theta)
\end{aligned}$$

Now consider

$$\begin{aligned}
\cos(3\theta) &= \frac{1}{2}(e^{3i\theta} + e^{-3i\theta}) \\
&= \frac{1}{2}[(e^{i\theta} + e^{-i\theta})^3 - 3e^{2i\theta}e^{-i\theta} - 3e^{i\theta}e^{-2i\theta}] \\
&= \frac{1}{2}[(e^{i\theta} + e^{-i\theta})^3 - 3(e^{i\theta} - e^{-i\theta})] \\
&= \frac{1}{2}[(2\cos\theta)^3 - 6\cos\theta] \\
&= 4\cos^3\theta - 3\cos\theta \\
\implies 4\cos^3\theta &= \cos(3\theta) + 3\cos\theta
\end{aligned}$$

3. **(a)** Find the Taylor series of $f(z) = \sum_{k=1}^{\infty}(-1)^{k+1}\sin(\frac{z}{k})$ about $z = 0$. What is its radius of convergence? **[8+4]**

   (b) What type of singularities (if any) possessed by (i) $f(z) = \frac{1}{z-3}$, (ii) $f(z) = \sinh(1/z)$, at $z = 0$ and $z = \infty$. **[4+4]**

Solution:
(a) The Taylor series of a function around $z = 0$ is given by

$$f(z) = \sum_{n=0}^{\infty} \frac{f^n(0)}{n!} z^n$$

Let's first compute $f^n(0)$

$$\begin{aligned}
f^n(z)|_{z=0} &= \frac{d^n}{dz^n} \sum_{k=1}^{\infty}(-1)^{k+1} \sin\left(\frac{z}{k}\right)\Big|_{z=0} \\
&= \sum_{k=1}^{\infty}(-1)^{k+1} \frac{d^n}{dz^n} \sin\left(\frac{z}{k}\right)\Big|_{z=0}
\end{aligned}$$

Clearly, this is 0 for even n. For odd n we get

$$f^n(z)|_{z=0} = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^n}(-1)^{\frac{n+3}{2}}$$

Hence the Taylor series around $z = 0$ is given by

$$f(z) = \sum_{n \text{ odd}} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^n}(-1)^{\frac{n+3}{2}} \frac{1}{n!} z^n$$

Note that the series is always less than the series for $\sin(z)$ and hence will always converge. Hence the radius of convergence is ∞.

(b) (i) $f(z) = \frac{1}{z-3}$ does not have a singularity at $z = 0$ or $z = \infty$. (ii) $f(z) = \sinh(1/z)$ has an essential singularity at $z = 0$ and no singularity at $z = \infty$.

4. Find α so that: **(a)** $\nabla\left(\frac{1}{r}\right) = -r^\alpha \vec{r}$, **(b)** $\nabla \cdot (r\vec{r}) = \alpha r$, **(c)** $\nabla \cdot (\nabla(r^3)) = \alpha r$, **(d)** $\nabla \times \nabla(r^\alpha) = \nabla \cdot \nabla \times (\sin(\alpha r)\vec{r})$ **[20]**

Solution:
(a)

$$\begin{aligned}
\nabla\left(\frac{1}{r}\right) &= -r^\alpha \vec{r} \\
\implies \frac{\partial}{\partial r}\left(\frac{1}{r}\right)\hat{r} &= -r^{\alpha-1}\hat{r} \\
\implies -\frac{1}{r^2}\hat{r} &= -r^{\alpha-1}\hat{r} \\
\implies \alpha &= -1
\end{aligned}$$

(b) This problem was removed from the test.

(c)

$$\begin{aligned}
\nabla \cdot (\nabla r^3) &= \alpha r \\
\implies \nabla \cdot (3r^2 \hat{r}) &= \alpha r \\
\implies \frac{\partial}{\partial r}(3r^2) &= \alpha r \\
\implies 6r &= \alpha r \\
\implies \alpha &= 6
\end{aligned}$$

(d)

$$\nabla \times (\nabla r^\alpha) = \nabla \cdot (\nabla \times (\sin(\alpha r))\vec{r})$$

Since Curl of a gradient is zero and divergence of curl is zero, both sides are 0 and the equation is true for any α.

5. Use Levi-Civita symbols to prove: **(a)** $\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$ **(b)** $n \times n$ square matrices A, B, C, D satisfy $\det(ABCD) = \det(DABC)$. **[10+10]**

Solution:
(a)

$$\begin{aligned}
\nabla \cdot (\vec{A} \times \vec{B}) &= \frac{\partial}{\partial x^i}(\epsilon_{ijk} A_j B_k) \\
&= \epsilon_{ijk} \frac{\partial A_j}{\partial x^i} B_k + \epsilon_{ijk} \frac{\partial B_k}{\partial x^i} A_j \\
&= \epsilon_{ijk} \frac{\partial A_j}{\partial x^i} B_k - \epsilon_{ijk} \frac{\partial B_j}{\partial x^i} A_k \\
&= \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})
\end{aligned}$$

(b) The required result follows trivially from $\det(AB) = \det(A)\det(B)$.

$$\begin{aligned}
\det(AB) &= \epsilon_{i_1 i_2 \ldots i_n} (AB)_{1i_1} (AB)_{2i_2} \ldots (AB)_{ni_n} \\
\det(AB) &= \epsilon_{i_1 i_2 \ldots i_n} A_{1k_1} B_{k_1 i_1} A_{2k_2} B_{k_2 i_2} \ldots A_{nk_n} B_{k_n i_n} \\
\det(AB) &= A_{1k_1} A_{2k_2} \ldots A_{nk_n} \epsilon_{i_1 i_2 \ldots i_n} B_{k_1 i_1} B_{k_2 i_2} \ldots B_{k_n i_n} \\
\det(AB) &= A_{1k_1} A_{2k_2} \ldots A_{nk_n} \det(B) \epsilon_{k_1 k_2 \ldots k_n} \\
\det(AB) &= \det(A)\det(B)
\end{aligned}$$